

Time Series

Applied Time Series Analysis

STA457 Lecture Notes

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Time Series Analysis

Models for Nonstationary Time Series and Model Specification

Integrated ARMA (or ARIMA) Models

- ① In practice, most often time series are nonstationary. In order to fit a stationary model, such as those discussed earlier, it is necessary to remove nonstationary source of variation.
- ② If the observed time series is nonstationary in the mean, then we can difference the series. More specifically, if the data exhibits no apparent deviations from stationarity and has a rapidly decreasing autocorrelation function, we shall seek a suitable ARMA process to represent the mean-correlated data. If not, then we shall first look for a transformation of the data which generates a new series with the above properties. This can frequently be achieved by differencing, leading us to consider the class of ARIMA (autoregressive-integrated moving average) processes.

- ③ If X_t is replaced by $\nabla^d X_t$ in the equation

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = W_t + \theta_1 W_{t-1} + \cdots + \theta_q W_{t-q} \quad (1)$$

then we have a model capable of describing certain types of nonstationary series.

- ④ Such a model is called an *integrated* model because the stationary model that is fitted to the differenced data has to be summed or *integrated* to provide a model for the original nonstationary data.

Definition

Stochastic process $\{X_t\}$ is said to be *intrinsically stationary* of integer order $d > 0$ if $\{X_t\}, \{\nabla X_t\}, \dots, \{\nabla^{d-1} X_t\}$ are nonstationary, but $\nabla^d X_t$ is a stationary process.

Remark

$$\begin{aligned}\nabla X_t &= (1 - B)X_t = X_t - X_{t-1} \\ \nabla^2 X_t &= \nabla(\nabla X_t) = (1 - B)^2 X_t = X_t - 2X_{t-1} + X_{t-2} \\ &\vdots \\ \nabla^d X_t &= (1 - B)^d X_t = \sum_{k=0}^d \binom{d}{k} (-1)^k B^k X_t = \sum_{k=0}^d \binom{d}{k} (-1)^k X_{t-k}\end{aligned}$$

Definition

If d is a nonnegative integer, then $\{X_t\}$ is said to be an ARIMA(p, d, q) process if

- ➊ $\{X_t\}$ is intrinsically stationary of order d
- ➋ $\{\nabla^d X_t\}$ is a causal ARMA(p, q) process

Remark

With $\{W_t\} \sim wn(0, \sigma_w^2)$, we can express the ARIMA(p, d, q) model as

$$\phi(B)(1 - B)^d X_t = \theta(B)W_t \quad (2)$$

If the process contains no autoregressive terms, we call it an integrated moving average and abbreviate the name to IMA(d, q). If no moving average terms are present, we denote the model as ARI(p, d).

Example

The simplest example of ARIMA process is $\text{ARIMA}(0,1,0)$:

Over-differencing

- ① While differencing a time series often seems to yield a series visually more amenable to modelling as a stationary process, over-differencing is a danger!

- ② If X_t is ARMA(p,q) already, and satisfies

$$\phi(B)X_t = \theta(B)W_t, \quad (3)$$

then one more differencing provides that

$$(1 - B)\phi(B)X_t = (1 - B)\theta(B)W_t \implies \phi(B)(1 - B)X_t = \theta(B)(1 - B)W_t; \quad (4)$$

i.e., one more difference of X_t , denoted by $Y_t = \nabla X_t$, satisfies

$$\phi(B)Y_t = \theta^*(B)W_t \quad (5)$$

where $\theta^*(z) = \theta(z)(1 - z)$. Note that $\theta^*(z)$ has a root on the unit circle. Thus, Y_t is a non-invertible ARMA(p,q+1) process.

- ③ Summary of evils of over-differencing:

- ① ARMA(p,q+1) model usually has more complex covariance structure than ARMA(p,q)
- ② ARMA(p,q+1) model has one more parameter to estimate than ARMA(p,q)

- ④ Unit root tests are often employed to determine if differencing is needed.

Dickey-Fuller Unit Root Tests

- 1 Consider the simple AR(1) model

$$X_t = \phi X_{t-1} + W_t \quad (6)$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$. The hypotheses of interest in unit root test are

$$H_0 : \phi = 1 \text{ (unit root in } \phi(z) = 0) \implies X_t \sim I(1) \quad (7a)$$

$$H_1 : |\phi| < 1 \implies X_t \sim I(0) \quad (7b)$$

- 2 Dickey-Fuller unit root test can assess the following cases:

1

$$H_0 : X_t = X_{t-1} + W_t \quad (8a)$$

$$H_1 : X_t = \phi X_{t-1} + W_t, \phi < 1 \quad (8b)$$

This is a test for a random walk against a stationary autoregressive process of order one (AR(1)).

2

$$H_0 : X_t = X_{t-1} + W_t \quad (9a)$$

$$H_1 : X_t = \delta + \phi X_{t-1} + W_t, \phi < 1 \quad (9b)$$

This is a test for a random walk against a stationary AR(1) with drift.

3

$$H_0 : X_t = X_{t-1} + W_t \quad (10a)$$

$$H_1 : X_t = \delta + \lambda t + \phi X_{t-1} + W_t, \phi < 1 \quad (10b)$$

This is a test for a random walk against a stationary AR(1) with drift and deterministic time trend.

Augmented Dickey-Fuller Unit Root Test

- ① The unit root test described in the previous slide are valid if the time series X_t is well characterized by an AR(1) with white noise errors.
- ② Many time series, however, have a more complicated dynamic structure than is captured by a simple AR(1) model.
- ③ One solution to overcome this issue is to augment the basic autoregressive unit root test to accommodate general ARMA(p,q) models with unknown orders. In this case, their test is referred to as the augmented Dickey-Fuller (ADF) test.
- ④ The ADF test tests the null hypothesis that a series X_t is $I(1)$ against the alternative that it is $I(0)$, assuming that the dynamics in the data have an ARMA structure.
- ⑤ The ADF test is based on estimating the test regression

$$x_t = \delta + \lambda t + \phi x_{t-1} + \sum_{j=1}^p \psi_j \Delta x_{t-j} + w_t \quad (11)$$

- ⑥ The lag length for the ADF test is usually determined using the following relation

$$p_{\max} = \lfloor 12 \cdot \left(\frac{T}{100} \right)^{0.25} \rfloor \quad (12)$$

Box-Jenkins Method

- ➊ The ARIMA(p, d, q) class of models as a broad class can describe many real time series.
- ➋ Model specification for ARIMA(p, d, q) models involves
 - ➊ Choosing appropriate values for p, d, and q;
 - ➋ Estimating the parameters (e.g., the ϕ s, θ s, and σ_w^2) of the ARIMA(p, d, q) model;
 - ➌ Checking model adequacy, and if necessary, improving the model.
- ➌ This process of iteratively proposing, checking, adjusting, and re-checking the model is known as the “Box-Jenkins method” for fitting time series models.

Determining the order of an AR Process (First Method)

- ① One approach for determining the order of $AR(p)$ is to fit AR processes of progressively higher order and calculate the residual sum of squares for each value of p .
- ② If we plot the residual values against p , we may then observe that the corresponding curve *flattens out* at some specific value of p where the addition of extra parameters gives little improvement in fit. Such a value of p would then be identified as the order of the autoregressive model.

Determining the order of an AR Process (Second Method)

① Another aid to determining the order of an AR process is the *partial autocorrelation function*, which is defined as follows:

① *When fitting an AR(p) model, the last coefficient ϕ_p measures the excess correlation at lag p which is NOT accounted for by an AR(p-1) model.*

② This value is called the *p*th *partial autocorrelation coefficient*, and when plotted against *p*, it gives the *partial ACF*.

③ The first partial autocorrelation coefficient π_1 is simply equal to ρ_1 .

④ For the case of AR(2) process, we can show that, the second partial autocorrelation coefficient π_2 is given by

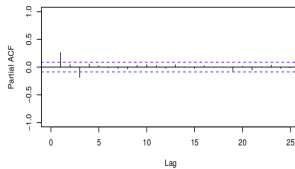
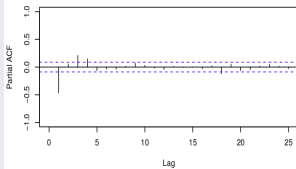
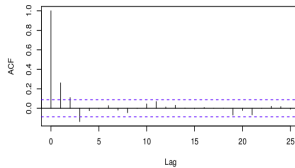
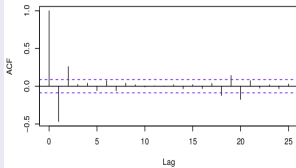
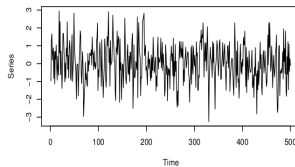
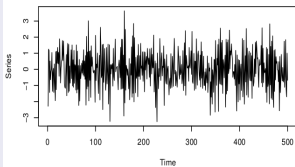
$$\frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

which is zero for AR(1) process as a result of having $\rho_2 = \rho_1^2$.

Sample Partial Autocorrelation Function for AR(p) and MA(q)

- 1 For AR process, it can be shown that the partial ACF, *cuts off* at lag p so that the *correct* order is assessed as that value of p beyond which the sample values of $\{\pi_j\}$ are not significantly different from zero.
- 2 In contrast, the partial ACF of an MA process will generally *die out slowly*. Thus, the partial ACF of AR model has *reverse* properties to that of MA process.

Time Plots, ACF and Partial ACF for Two AR Processes



Sample Partial Autoorrelation

Sample Partial Autoorrelation

Sample Partial Autoorrelation

Sample Partial Autoorrelation

Model Identification Tools

- ④ Model building is a key element of most statistical work. To formulate an appropriate model, the analyst would typically look at the *time plot* of the data and at various diagnostic tools. The two standard tools, which are used to identify an appropriate model for a given *stationary* time series, are the *sample ACF* and the *sample partial ACF*.
- ⑤ This choice is often made subjectively, using the analyst's experience to match an appropriate model to the observed characteristics. However, various additional diagnostic tools are also available which are quantitative measures and make the task of comparing different models more systematic and objective.
- ⑥ "What criterion should we use to select a model in the best way?" It is not sensible to simply choose a model to give the best fit by minimizing the residual sum of squares, as the latter will generally decrease as the number of parameters is increased regardless of whether the additional complexity is really worthwhile.
- ⑦ There is an alternative fit statistic, called adjusted- R^2 , which makes some attempt to take account of the *number* of parameters fitted.

Model Identification Tools

- ➊ More sophisticated model selection criteria are also available. *Akaike's Information Criterion* (AIC) is the most commonly used and is given approximately by

$$AIC = -2 \log(\text{max. likelihood}) + 2r \quad (13)$$

where r denotes the number of independent parameters that are fitted for the model being assessed.

- ➋ The AIC essentially chooses the model with the best fit, as measured by the likelihood function, subject to a penalty term, in order to prevent overfitting since we know that overfitting increases with the number of parameters in the model.
- ➌ It is, however, extremely important to note that the AIC is *biased* for *small samples*.
- ➍ For this reason, a bias-corrected version of AIC is often recommended which is formulated as follows:

$$AIC_C = -2 \log(\text{max. likelihood}) + \frac{2rN}{N - r - 1}. \quad (14)$$

Model Identification Tools

- ① Another widely used criterion is the *Bayesian Information Criterion* (BIC) which is formulated as follows:

$$BIC = -2 \log(\text{max. likelihood})(r + \log N). \quad (15)$$

- ② Thus, it penalizes the addition of extra parameters more severely than the AIC.
- ③ *So, in some sense, AIC overestimates the number of parameters in the model, whereas BIC underestimates this quantity.*
- ④ Other popular model selection criteria are as follows:
 - ① Schwartz's Bayesian Criterion: $SBC = -2 \log(\text{max. likelihood})(r \log N)$
 - ② Autoregressive Transfer Function Criterion (CAT)
 - ③ Akaike's Final Prediction Error (FPE)

Extended Autocorrelation Function

- ❶ For an ARMA(p, q) model, the true ACF and true PACF both have infinitely many nonzero values.
- ❷ Neither the true ACF nor the true PACF will cut off entirely after a certain number of lags.
- ❸ So it is hard to determine the correct orders of an ARMA(p, q) model simply by using the ACF and PACF.
- ❹ The extended autocorrelation function (EACF) is one method proposed to assess the orders of a ARMA(p, q) model.
- ❺ The EACF table for an ARMA(p, q) process should theoretically have a triangular pattern of zeroes with the top-left zero occurring in the p -th row and q -th column (with the row and column labels both starting from 0).
- ❻ (In reality, the sample EACF table will not be as clear-cut as the examples that follow, since the sample EACF values have sampling variability.)

Theoretical EACF Table for an ARMA(1, 1) Process

AR/MA	0	1	2	3	4	5	6	7	8	9	10
0	x	x	x	x	x	x	x	x	x	x	x
1	x	0	0	0	0	0	0	0	0	0	0
2	x	x	0	0	0	0	0	0	0	0	0
3	x	x	x	0	0	0	0	0	0	0	0
4	x	x	x	x	0	0	0	0	0	0	0
5	x	x	x	x	x	0	0	0	0	0	0
6	x	x	x	x	x	x	0	0	0	0	0
7	x	x	x	x	x	x	x	0	0	0	0

Theoretical EACF Table for an ARMA(2, 3) Process

AR/MA	0	1	2	3	4	5	6	7	8	9	10
0	x	x	x	x	x	x	x	x	x	x	x
1	x	x	x	x	x	x	x	x	x	x	x
2	x	x	x	0	0	0	0	0	0	0	0
3	x	x	x	x	0	0	0	0	0	0	0
4	x	x	x	x	x	0	0	0	0	0	0
5	x	x	x	x	x	x	0	0	0	0	0
6	x	x	x	x	x	x	x	0	0	0	0
7	x	x	x	x	x	x	x	x	0	0	0